

We are solving hashcrack, an easy-level cryptography challenge on cylab security academy. In this challenge, we must crack a secret message leaked by a server due to the admin using weakly hashed passwords.

I start off by connecting to the challenge instance via netcat and it asks for a password given a hash:

```
(hackingmastermind@ctfdestroyingMACHINE)-[~]
$ nc verbal-sleep.picoctf.net 64655
Welcome!! Looking For the Secret?
hashcrack
We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: █
```

I knew I had to crack the hash to know the password, so I thought of using hashcat, a very well-known password cracking tool. Hashcat requires me to know the hashing algorithm used by the server to hash the password, so I used an online hash identifier tool to determine what it could be and it turns out to be either md5, md2, or md4 (I had a feeling it was most likely md5 since that hashing algorithm comes up a lot in these kinds of challenges):



Now that I knew what hashing algorithm was being used, I was ready to use hashcat. To run hashcat, I need to specify the hashing algorithm used. Hashcat assigns numbers to each hashing algorithm. The number of the md5 algorithm is 0. I specify the attack mode as 0 using the -a option (use passwords from the dictionary file one after the other), the hash file (the file I put my hash in), and a dictionary file. However, when I tried to run it, it failed since I was missing some software:

```

(hackingmastermind@ctfdestroyingMACHINE)-[~]
$ hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting

clGetPlatformIDs(): CL_PLATFORM_NOT_FOUND_KHR

ATTENTION! No OpenCL, HIP or CUDA compatible platform found.

You are probably missing the OpenCL, CUDA or HIP runtime installation.

* AMD GPUs on Linux require this driver:
  "AMD Radeon Software for Linux" with "ROCm"
* Intel and AMD CPUs require this runtime:
  "Intel CPU Runtime for OpenCL" or PoCL
* Intel GPUs require this driver:
  "Intel Graphics Compute Runtime" aka NEO
* NVIDIA GPUs require this runtime and driver:
  "NVIDIA CUDA Toolkit" (both runtime and driver included)

Started: Fri Sep  4 09:33:42 2026
Stopped: Fri Sep  4 09:33:42 2026

```

I tried to switch to john. John is a very well known password cracking tool as well. To use john, you specify the hash algorithm used (you don't necessarily need to but it makes things a lot easier), the dictionary file to attack the hash, and the hash file. However, john was taking too long to run:

```

(hackingmastermind@ctfdestroyingMACHINE)-[~]
$ john --format=raw-md5 /usr/share/wordlists/rockyou.txt hash.txt
Warning: invalid UTF-8 seen reading /usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 53 password hashes with no different salts (Raw-MD5 [MD5 256/256 AVX2
8x3])
Remaining 52 password hashes with no different salts
Warning: no OpenMP support for this hash type, consider --fork=2
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
Proceeding with incremental:ASCII

```

I finally decided to use an online hash cracking tool. The one I used was CrackStation. I pasted the hash and found the plaintext:



## Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3

I'm not a robot

reCAPTCHA

Crack Hashes

**Supports:** LM, NTLM, md2, md4, md5, md5(md5\_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1\_bin)), QubesV3.1BackupDefaults

| hashes: 1                                | cracked: 1 | partial: 0 | not found: 0 | invalid: 0 |
|------------------------------------------|------------|------------|--------------|------------|
| Hash                                     | Type       | Result     |              |            |
| b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3 | sha1       | letmein    |              |            |

**Color Codes:** Green: Exact match, Yellow: Partial match — the hash column shows what that word really hashes to, with the part yours matched highlighted, Red: Not found.

I then typed it into the program and was yet again greeted with another hash to crack:

```
(hackingmastermind@ctfdestroyingMACHINE)-[~]
$ nc verbal-sleep.picoctf.net 51961
Welcome!! Looking For the Secret?
No Crack
We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: password123
Correct! You've cracked the MD5 hash with no secret found!

Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3
c52ab3ce3
Enter the password for the identified hash: letmein
Correct! You've cracked the SHA-1 hash with no secret found!

Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f992
8310ad30a8d88697745
Enter the password for the identified hash: 
```

I use Crackstation again to crack the hash:

## Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745

☐

I'm not a robot

reCAPTCHA

Crack Hashes

**Supports:** LM, NTLM, md2, md4, md5, md5(md5\_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1\_bin)), QubesV3.1BackupDefaults

| hashes: 1                                                        | cracked: 1 | partial: 0 | not found: 0 | invalid: 0 |
|------------------------------------------------------------------|------------|------------|--------------|------------|
| Hash                                                             | Type       | Result     |              |            |
| 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745 | sha256     | qwerty098  |              |            |

**Color Codes:** Green Exact match, Yellow Partial match — the hash column shows what that word really hashes to, with the part yours matched highlighted, Red Not

I type it into the program and finally get the flag:

```

Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f992
8310ad30a8d88697745
Enter the password for the identified hash: qwerty098
Correct! You've cracked the SHA-256 hash with a secret found.
The flag is: picoCTF{UseStr0nG_h@shEs_&PaSswDs!_6965e43b}

(hackingmastermind@ctfdestroyingMACHINE)~[~]
$

```

flag: picoCTF{UseStr0nG\_h@shEs\_&PaSswDs!\_6965e43b}